

What It Is

CIE Copilot is a learning and assessment platform for Cambridge International Education subjects. This repo centers on RAG search/Q&A, community workflows, recommendations, learning paths, and Phase1 marking.

Who It Is For

Primary persona: CIE A Level students, especially Mathematics, Further Mathematics, and Physics learners.

What It Does

- Browse subjects, papers, topics, and concept content.
- Search syllabus-aligned material with RAG.
- Run Ask AI and AI tutoring chat flows.
- Generate recommendations and learning paths.
- Support community questions, answers, profiles, badges, and notifications.
- Track error-book entries and server-side marking via evaluate-v1.

How To Run

- Prereqs: Node >= 20, npm >= 9, Python >= 3.11, PostgreSQL/Supabase access.
- Install deps: `npm ci` and `python -m pip install -r requirements.txt`.
- Copy `.env.example` to `.env`.
- Start frontend: `npm run dev`.
- Start API: `npm run api:dev`.
- Frontend vars are documented in `.env.example.frontend`; full one-command bootstrap is Not found in repo.

How It Works

- Frontend: React + Vite SPA bootstraps in `src/main.jsx` and routes from `src/App.jsx`.
- App state uses React Query plus Theme, Auth, and AI providers.
- Data access is hybrid: some client flows use Supabase directly; others call `/api/*` clients for AI, RAG, and error-book features.
- `api/index.js` is the Node HTTP gateway for CORS, auth, rate limiting, route lookup, and handler loading.
- Domain modules live under `api/ai`, `api/rag`, `api/recommendations`, `api/community`, `api/error-book`, `api/marketing`, and `api/users`.
- Storage is Supabase on both sides: anon access in the browser and service-role access on the server.
- Typical flow: `page -> service/client -> Supabase or /api/* -> handler -> tables/RPC/ledger -> JSON back to UI`.

Repo Evidence

README.md; `src/main.jsx`; `src/App.jsx`;
`src/utlis/supabase.js`; `api/index.js`;
`api/_runtime/route-registry.js`; `api/lib/supabase/client.js`